TOPOSCOUT

GNSS SURVEY PLANNING SYSTEM

SITE INTELLIGENCE REPORT

REPORT SG-6Z2S9X
Generated July 13, 2026
Data current as of report date

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES	ELEVATION	STATE PLANE	FEMA ZONE	DECLINATION
38.3459°N 122.3047°W WGS84	**66 ft** NAVD88 · MSL	**CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83)**	**Zone X** 06055C0510FF	**13.0° E** WMM 2026

APN **038050009000** | ZONING **AP** | ACRES **0.481 ac**

BEST GNSS WINDOW **Mon, Jul 13 · 6:00 AM – 10:00 AM** | **PDOP 0.99 · 34 sv** | NEAREST CORS **P261 · 14.1 mi ✓**

REPORT CONTENTS

SCAN FOR KMZ
Site file · Google Earth
↓ Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 13, 2026

ASSESSOR
Napa County Assessor · 1127 First St, Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

TOPOSCOUT
★ A SITE INTELLIGENCE REPORT ★

SITE & SIGNAL REPORT

COORDINATES	ELEVATION	MAGNETIC DECLINATION	FEMA FLOOD ZONE
38.3459, -122.3047	**66 ft**	**13.0° E**	**X**
WGS84 / NAD83	Above MSL (approx)	Compass reads 13.0° too far W	AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX
4.7

AVG KP
2.4

SOLAR FLUX F10.7
138 sfu

TEC (VTEC) ⚠ DEGRADED
16.5 TECU
-5.3 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Mon, Jul 13	0.99	6:00 AM – 10:00 AM	34	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location — orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

🔥 WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 13, 2026) · Conditions may change — verify before fieldwork

🛰 GPS CONSTELLATION HEALTH

⚠ 1 Satellite Currently Unhealthy — PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

🌿 Land Cover & Multipath Risk

SKY OBSTRUCTION

MULTIPATH RISK

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–0.8 across survey window

PDOP remains ≤ 0.8 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Monday, Jul 13

Overcast | 💨 9 mph | **GO**

High 93°F | Low 58°F | Wind max 9 mph | Gusts 12 mph | Precip 0.00" | Sunrise 5:56 AM | Sunset 8:33 PM

🌘 Waning Crescent 2%

GNSS: Avg PDOP **0.99** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.94, 34 sats avg)

GNSS / DOP — HOURLY

	PDOP	SATS	RATING
6:00 AM	0.84	36 sv	IDEAL
7:00 AM	0.95	34 sv	IDEAL
8:00 AM	0.95	33 sv	IDEAL
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.01	29 sv	GOOD
11:00 AM	1.11	25 sv	GOOD
12:00 PM	1.03	26 sv	GOOD
1:00 PM	1.08	26 sv	GOOD
2:00 PM	0.93	32 sv	IDEAL
3:00 PM	0.97	33 sv	IDEAL
4:00 PM	0.98	33 sv	IDEAL
5:00 PM	0.99	33 sv	IDEAL
6:00 PM	1.02	30 sv	GOOD
7:00 PM	0.99	32 sv	IDEAL

⛅ **GO — GOOD CONDITIONS**

Partly cloudy | Wind: 9.4 mph (gusts 11.6 mph) | Precip: 0" | Temp: 58.1°–93.2°F

🚁 DRONE WIND ANALYSIS

✅ **GO** Peak: 9.4/11.6 mph S

☀ Sunrise: **5:56 AM** ☾ Sunset: **8:33 PM** fly window clipped to daylight

✅ Best flight window: **6:00 AM – 9:00 PM (15 hrs)**

🔄 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph ⚡=turbulence	DIR
5 AM	2/3 mph	SE
6 AM	3/4 mph	ESE
7 AM	1/2 mph	ESE
8 AM	2/3 mph	SSE
9 AM	4/5 mph	S
10 AM	3/3 mph	SSE
11 AM	5/6 mph	S
12 PM	4/6 mph	S
1 PM	7/7 mph	S
2 PM	9/10 mph	S
3 PM	7/8 mph	S
4 PM	8/9 mph	SSW
5 PM	9/11 mph	SSW
6 PM	9/11 mph	SSW
7 PM	9/12 mph	S
8 PM	8/11 mph	S

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS A** ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
✈ TFR STATUS ✅ **NO TFRs WITHIN 30 MI** Check B4UFLY ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

AI FIELD ANALYSIS — CLAUDE HAIKU 4.5

FIELD PLANNING REPORT

Napa County, CA | 38.3459, -122.3047

Survey Date: Monday, July 13, 2026

Report Generated: Monday, July 13, 2026

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VERDICT: GO

Primary Reason: Excellent PDOP geometry (0.99 average, 0.94 in optimal window) combined with acceptable weather and moderate ionospheric conditions support field operations. Single unhealthy GPS satellite (PRN 11) does not materially degrade constellation.

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BEST SURVEY WINDOW: 06:00 – 10:00 (Local Time)

Rationale: PDOP 0.94 with 34-satellite availability (multi-constellation). This 4-hour window provides optimal geometric strength for both rapid static and kinematic work. Peak sun angle supports clear line-of-sight to all visible satellites.

==

BEST 4-HOUR STATIC GNSS WINDOW: 06:30 – 10:30

Recommended session: 06:30 – 10:30 (or split into two 2-hour sessions if multiple points required). PDOP remains below 1.0 throughout. At moderate Kp 4.7, ionospheric delay is manageable; recommend minimum 30-minute occupations per point to average residual TEC noise (note: TEC estimate flagged; treat conservatively). Post-process with multi-constellation solution to maximize redundancy given single GPS satellite outage.

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DRONE OPERATIONS ASSESSMENT: APPROVED

Wind max 9 mph well within safe operating limits for most small UAS platforms. Visual line-of-sight operations recommended through 10:00 local time due to overcast conditions; operations feasible but avoid extended high-altitude hover after 11:00 if cumulus development increases. No fire proximity risk.

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FIELD SAFETY WARNINGS

— Moderate geomagnetic activity (Kp 4.7): If site includes sensitive magnetometer work, expect possible ±2–5 nT baseline shift; note observation time.

— TEC data quality flagged: Ionospheric estimates unreliable; rely on dual-frequency observations where possible.

— Overcast sky: Bring sun protection despite cloud cover; UV index remains elevated in July.

— Standard site hazards: Low multipath risk; elevation mask 13° sufficient for terrain clearance.

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SUMMARY

Monday, July 13 is operationally GO. Geometry is excellent (PDOP 0.99–0.94) and weather is benign (overcast, 9 mph winds, no rain). Moderate ionospheric activity (Kp 4.7, F10.7 138 sfu) introduces minor delay but is well within static correction capability; extend occupations to 30 minutes minimum per point. GPS PRN 11 is unhealthy but multi-constellation strategy (GLONASS + Galileo + BeiDou) maintains 34-satellite strength. Execute primary survey between 06:00–10:00 local time for best results. Drone operations approved. No wildfire or air quality constraints.

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk

1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor

1127 First St, Napa, CA 94559

CORS STATIONS — WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
NGS ↗	**P261**	HUNTERHILLCN2004	14.1 mi
NGS ↗	**P198**	PETALUMAIRCN2004	17.5 mi
NGS ↗	**OHLN**	OHLONE PARK	23.5 mi
NGS ↗	**P196**	MEACHUMLFLCN2006	24.0 mi
NGS ↗	**CASR**	SANTA ROSA CA	24.9 mi
NGS ↗	**P267**	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS — WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗	**JT9010**	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9018**	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9009**	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9017**	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9034**	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
Datasheet ↗	**JT9012**	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
Datasheet ↗	**JT9043**	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X — AREA OF MINIMAL FLOOD HAZARD

Panel: **06055C0510FF**
Effective: Sep 29, 2010
View FEMA Map ↗

DATA COLLECTOR IMPORT — NGS BENCHMARKS

↓ Download .txt File

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────────────────────────────────────────────────────────────────────────
NGS BENCHMARK CONTROL POINTS — Copy / Paste to Data Collector
Datum: NAD83  |  Vertical: NAVD88  |  Source: NGS NDE API
────────────────────────────────────────────────────────────────────────
PID      NAME                    LAT           LON           ELEV_FT  ORDER
────────────────────────────────────────────────────────────────────────
JT9010   13                        38.348333    -122.302778       62.7
JT9018   13 E                      38.348333    -122.302778       59.0
JT9009   13                        38.346667    -122.301111       58.7
JT9017   13 D                      38.345000    -122.301111       58.7
JT9034   20 A                      38.345000    -122.309444       68.0
JT9012   13                        38.351667    -122.304444       62.5
JT9043   25 A                      38.340000    -122.302778       61.4
JT9016   13 D                      38.341667    -122.297778       59.7
────────────────────────────────────────────────────────────────────────
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CALIFORNIA RESERVOIRS (DWR/CDEC) — within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.93 ft (-9 ft)	706 ft	36K af	**67%**
Folsom Lake	66.5 mi	448.12 ft (-18 ft)	466 ft	787K af	**81%**
Lake Oroville	93.4 mi	861.09 ft (-39 ft)	900 ft	2,857K af	**81%**
New Melones Lake	100.1 mi	1,020.09 ft (-68 ft)	1,088 ft	1,663K af	**69%**
Black Butte Lake	101.2 mi	—	228 ft	74K af	**54%**

USBR RESERVOIRS (Bureau of Reclamation) — within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP

ARMY CORPS RESERVOIRS (USACE) — within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.9 ft	695.01 ft	—	**100%**
Camanche	89 mi	230.3 ft	221.86 ft	—	**104%**
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	**101%**
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	**107%**
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	**100%**
Don Pedro	137 mi	—	N/R	—	**—**

USGS GROUNDWATER WELLS — within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL
004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

⬡ AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Napa Valley Coastal Ranges aquifer (N100CACSTL) consists of Tertiary-age sandstone, siltstone, and shale deposited in marine and fluvial environments during the Oligocene to Pliocene. This moderately consolidated sedimentary sequence underlies much of Napa County and yields water through intergranular porosity and fracture flow. Water availability at this site is moderate; saturated thickness varies with local geology, and water levels reflect both seasonal recharge and long-term basin stress.

FORMATION

Name	*Napa Valley Coastal Ranges Tertiary Sedimentary Complex*
Age / Rock	*Oligocene to Pliocene (23–2.6 Ma); sandstone, siltstone, shale, and minor conglomerate.*
Extent	*800–1200 sq mi (est. USGS/DWR)*

DEPTH & RECHARGE

Depth Range	*51–295 ft (5–30 floors)*
Typical Depth	*100–150 ft (10–15 floors)*
Recharge Source	*Precipitation on outcrop areas in Coastal Ranges; lateral flow from higher-elevation recharge zones*
Transit Time	*20–50 years to 100 ft depth (est. USGS/DWR)*

WATER QUALITY

pH	*est. 7.0–7.6 (est. USGS/DWR)*
Hardness	*est. 150–350 mg/L CaCO3 (est. USGS/DWR)*
TDS	*est. 250–500 mg/L (est. USGS/DWR)*

Notes	*No sample data available — typical for formation. Expected to be fresh to slightly brackish depending on proximity to marine paleoenvironments and depth.*

TREND

Trend	*Mixed: shallow wells (9.90–41.90 ft BGS, 1977) show long-term decline; deepest well (87.71 ft BGS, 2014) indicates continued drawdown. One shallow measurement (21.86 ft BGS, 2022) suggests some recovery or seasonal variation.*
Source	*USGS NWIS well record measurements 1977–2022*
Seasonal Swing	*est. 10–25 ft (seasonal recharge and pumping cycles)*

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

⬡ AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The Coast Ranges near 38.3459, -122.3047 comprise consolidated Tertiary marine sediments and Mesozoic metamorphic bedrock with minimal clay aquitard sequences, resulting in naturally good drainage and limited compressible interbeds. The site lies outside the state's critically overdrafted groundwater basins where subsidence occurs, and the N100CACSTL aquifer code indicates a relatively stable hydrogeologic setting without chronic over-pumping. For surveyors working in this area, benchmark elevations should remain stable and suitable for control without subsidence corrections.

SUBSIDENCE ASSESSMENT

Susceptibility	*Low*
Primary Mechanism	*None - consolidated terrain*
Annual Rate	*<1 mm/yr (est. USGS)*
Historic Loss	*None documented*
Aquifer Compaction	*None - aquifer not over-pumped*
SGMA Status	*Not applicable - not in SGMA basin*

BENCHMARK IMPLICATIONS

Monument Risk	*Low - monuments stable*
Elevation Validity	*Good - no active subsidence*
GNSS Recommendation	*Standard benchmark verification adequate*

SURVEY ACTION

Required Action	*No special action required*
Dataset	*CA DWR SGMA Bulletin 118 2024.1*
Coverage	*California critically overdrafted basins only - full US coverage pending*

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	92 mph
Ground Snow Load	5.8 psf
Winter Wind (W2)	0.35
Ice Thickness	None
Frost Depth	**12 in minimum**

Est. from IECC Zone 3C — AFI data unavailable

Verify with local building official before construction

Source: ASCE 7-22 Hazard Tool · gis.asce.org

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

⬡ AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This marine climate stays mild year-round with moderate moisture, so you can prioritize natural ventilation and daylighting without major heating or cooling loads—your envelope doesn't need extreme insulation, but air sealing matters to manage the persistent coastal dampness and prevent condensation issues. Orient primary glazing south and east to capture winter sun and passive heat, but minimize west-facing glass to avoid afternoon overheating, and keep glazing back from ocean-facing facades where salt spray accelerates deterioration. The single most effective passive strategy here is operable windows positioned for cross-ventilation; you can rely on consistent coastal breezes to naturally cool the home most of the year, dramatically reducing mechanical cooling needs.

⬡ AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This marine coastal site in IECC Zone 3C presents moderate wind exposure and minimal snow loading, with well-drained gravelly loam soils anticipated to support conventional spread foundations. Seismic parameters are currently unavailable and may warrant geotechnical investigation to establish site class confirmation and assess any foundation design implications. The 12-inch frost depth and excellent drainage suggest standard foundation detailing may be adequate, though local verification is recommended.

FOUNDATION

Type	*Conventional spread footing anticipated*
Basis	*Well-drained Coombs gravelly loam with good bearing characteristics suggested; seismic parameters unavailable — consult licensed engineer and geotechnical professional to confirm site class and establish foundation depth requirements relative to 12-inch frost line*

STRUCTURAL IMPLICATIONS

Seismic	*Seismic Design Category and response spectra currently null — geotechnical and structural engineer consultation recommended to establish SDC and confirm whether standard or special detailing may be warranted*
Wind Exposure	*Exposure C (open terrain) — 92 mph 3-second gust anticipated per ASCE 7-22 Risk Category II; site conditions suggest coastal influence consistent with marine climate zone*

Wind	*Standard wind load design anticipated for Risk Category II — no hurricane exposure indicated; consult structural engineer to confirm exposure category and load coefficients relative to final site plan*
Snow	*Light to moderate snow loading anticipated (5.8 psf ground load, W2=0.35) — standard roof design likely adequate; confirm with structural engineer relative to roof pitch and exposure*

INVESTIGATIONS & INSPECTIONS

Geotech Required	*Yes — geotechnical investigation recommended to confirm Site Class D assumption, establish bearing capacity, verify frost depth with local building official, and screen for any subsurface conditions*
Liquefaction Screen	*Screening recommended — Site Class D and groundwater depth unknown; consult geotechnical engineer to evaluate liquefaction potential and recommend mitigation if indicated*
Special Inspections	*Not anticipated at this planning stage; requirements depend on SDC determination — consult structural and geotechnical engineers once seismic parameters and site class are confirmed*
Code Edition	*ASCE 7-22 / IBC 2021 (planning reference only)*
Site Class Note	*Site Class D assumed for planning based on soil description; geotechnical investigation recommended to confirm classification, establish shear wave velocity if seismic design required, and verify drainage and frost conditions*

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes

SERIES: Coombs

DRAINAGE CLASS: Well drained

HYDRO GROUP: C

SOIL ORDER: Alfisols

SUBORDER: Xeralfs

⬡ AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is an Alfisol (Xeralf suborder) formed in mixed alluvial and colluvial materials on gently rolling terrain in the interior Coast Ranges or similar Mediterranean climates. This well-drained soil features a gravelly loam surface layer overlying a clay-enriched substratum, typical of soils developed under xeric (dry summer) conditions. Surveyors and developers should expect moderate gravel content, good trafficability on slopes under 5%, and stable bearing conditions, though the clay substratum requires attention during grading and fill placement.

Mixed alluvial and colluvial parent material on foothill slopes under oak-grassland and chaparral vegetation.

DRAINAGE

Class	*Well drained*
Seasonal Saturation	*None within 60 inches*
Flooding Risk	*Minimal on 2–5% slopes*
Ponding Risk	*None*

ENGINEERING

Bearing Capacity	*3,500–4,500 psf (shallow foundations)*
Rating	*Good*
Shrink-Swell	*Moderate — clay-enriched substratum; monitor moisture fluctuation*
Frost Action	*Low*
Cut Suitability	*Good (gravelly loam material; stable slopes)*
Fill Suitability	*Fair (clay substratum — requires moisture control and compaction verification)*

SEPTIC

Suitability	*Conditional*
Notes	*Perc test required — clay loam/clay substratum at 12–30 inches may restrict percolation; design must account for slower drainage rates typical of Hydrologic Group C soils*

HAZARDS

Excavation	*Moderate — 15–30% gravel content; equipment efficiency reduced; possible boulder strikes*
Expansive Clay	*Low to Moderate — monitor clay substratum during wet season; limited expansion risk in xeric climate but possible near groundwater contact*
Erosion	*Low on stable 2–5% slopes; gully initiation risk in disturbed or stripped areas; recommend erosion control during grading*
Other	*None*

Data sources: California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdmdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

Water levels current as of report run: Jul 13, 2026, 3:44 AM UTC

* Live pool elevation not reported by this reservoir — full pool design elevation shown

🛢 OIL & GAS WELLS California — within 10mi radius

8	0	7	By status:
total	active	plugged	**7** Plugged **1** Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	**Plugged**	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	**Idle**	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	**Plugged**	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation dry hole located 7.2 miles away, which is well beyond any actionable distance threshold. No active producers, injection wells, or other well types exist within relevant planning distances.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

⛏ MINES & MINERAL DEPOSITS USGS MRDS — within ~35mi radius

200 total in search area

10 within 5 mi

108 within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	**Occurrence**	Sand/Gravel	1.6 mi
Basalt Rock Co	**Producer**	Clay	2 mi
Cicero Pumice Quarry	**Occurrence**	PUM	2.9 mi
Mount George King Pumice Deposit.	**Occurrence**	PUM	2.9 mi
Basalt Rock Co.	**Occurrence**	DIT	3.1 mi
Mt George Pumice Deposit	**Producer**	PUM	3.1 mi
Wilson Pumice Quarry	**Occurrence**	PUM	3.7 mi
Unknown	**Occurrence**	Mercury	4.4 mi
Silva Pumice Quarry	**Occurrence**	PUM	4.7 mi
Silva Pumice Quarry	**Producer**	PUM	5 mi
Walker Pumice Quarry	**Occurrence**	PUM	5.1 mi
Walker Pumice Quarry	**Producer**	PUM	5.1 mi
Mountain	**Occurrence**	Mercury	5.4 mi
Unknown	**Occurrence**	Mercury	7.6 mi
Unknown	**Occurrence**	Stone/Crushed	7.8 mi
Unnamed Location	**Past Producer**	Sand/Gravel	8.5 mi
Unnamed Location	**Past Producer**	Sand/Gravel	8.5 mi
Unnamed Prospect	**Occurrence**	Chromium	8.8 mi
Unnamed Location	**Occurrence**	Chromium	9 mi
Unknown	**Occurrence**	Chromium	9.1 mi
Bella Oak	**Past Producer**	Mercury	9.4 mi
Unknown	**Occurrence**	MG	9.7 mi
Serres Pit	**Past Producer**	Sand/Gravel	10.4 mi
Unnamed Location	**Unknown**	Sand/Gravel	10.5 mi
Conn Valley	**Occurrence**	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Mineral Resource Review: 38.3459°N, 122.3047°W Three active producers operate within 5 miles of the site. Basalt Rock Co (2 mi) produces clay and operates with typical crushing/processing noise; the same operator has a separate occurrence at 3.1 mi. Silva Pumice Quarry (5 mi) and Mt George Pumice Deposit (3.1 mi) are active pumice producers. All three operations will generate heavy truck traffic on access roads during operating hours, with dust and potential blasting noise typical of aggregate and pumice extraction.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 Endangered

6 Threatened

12 Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	*SCIENTIFIC NAME*	STATUS
California tiger Salamander	*Ambystoma californiense*	**Endangered**
Conservancy fairy shrimp	*Branchinecta conservatio*	**Endangered**
Contra Costa goldfields	*Lasthenia conjugens*	**Endangered**
Soft bird's-beak	*Cordylanthus mollis ssp. mollis*	**Endangered**
Suisun thistle	*Cirsium hydrophilum var. hydrophilum*	**Endangered**
Vernal pool tadpole shrimp	*Lepidurus packardi*	**Endangered**
Alameda whipsnake (=striped racer)	*Masticophis lateralis euryxanthus*	**Threatened**
California red-legged frog	*Rana draytonii*	**Threatened**
Delta smelt	*Hypomesus transpacificus*	**Threatened**
Northern spotted owl	*Strix occidentalis caurina*	**Threatened**
Vernal pool fairy shrimp	*Branchinecta lynchi*	**Threatened**
western snowy plover	*Charadrius nivosus nivosus*	**Threatened**

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Review Summary The proposed project at 38.3459, -122.3047 triggers Section 7 consultation requirements if federal permits, funding, or approvals are involved, given the presence of six federally endangered and six threatened species with critical habitat within 25 miles. The most planning-relevant species include California tiger salamander and vernal pool species (tadpole shrimp, fairy shrimp), which indicate potential seasonal wetland impacts, and California red-legged frog, which requires aquatic and riparian habitat assessment. Site reconnaissance should prioritize identifying vernal pools, seasonal wetlands, and suitable upland migration corridors to determine project overlap with species habitat and critical habitat designations. Recommend a Biological Assessment and Habitat Assessment Report prepared by qualified environmental professionals to evaluate species presence, project impacts, and mitigation feasibility prior to permitting. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57	5051	2026-07-04
Unique Species	Total Observations	Most Recent

COMMON NAME	*SCIENTIFIC NAME*	LAST OBS
American black nightshade	*Solanum americanum*	2026-05-29
Bearded Clover	*Trifolium barbigerum*	2026-05-15
bird's foot cliffbrake	*Pellaea mucronata*	2026-04-28
Blow Wives	*Achyrachaena mollis*	2026-05-20
blue oak	*Quercus douglasii*	2026-05-09
Bolander's Phacelia	*Phacelia bolanderi*	2026-04-27
box elder	*Acer negundo*	2026-05-18
Braun's Giant Horsetail	*Equisetum telmateia braunii*	2026-06-22
California bay	*Umbellularia californica*	2026-06-24
California beeplant	*Scrophularia californica*	2026-05-20
California black oak	*Quercus kelloggii*	2026-05-15
California buckeye	*Aesculus californica*	2026-06-02
California Dutchman's Pipe	*Aristolochia californica*	2026-05-26
California milkwort	*Rhinotropis californica*	2026-05-15
California mugwort	*Artemisia douglasiana*	2026-06-02
California poppy	*Eschscholzia californica*	2026-06-24
California Wild Rose	*Rosa californica*	2026-05-09
Canyon liveforever	*Dudleya cymosa*	2026-05-30
-	*Castilleja densiflora densiflora*	2026-05-09
clay mariposa lily	*Calochortus argillosus*	2026-05-15
coast live oak	*Quercus agrifolia*	2026-05-13
coastal woodfern	*Dryopteris arguta*	2026-05-20
common selfheal	*Prunella vulgaris*	2026-06-22
Diogenes' lantern	*Calochortus amabilis*	2026-05-15
distant phacelia	*Phacelia distans*	2026-05-15
-	*Downingia concolor concolor*	2026-05-15
Dwarf Brodiaea	*Brodiaea terrestris*	2026-05-15
Elegant Clarkia	*Clarkia unguiculata*	2026-05-16
goldback fern	*Pentagramma triangularis*	2026-05-20
Ithuriel's Spear	*Triteleia laxa*	2026-06-15

COMMON NAME	*SCIENTIFIC NAME*	LAST OBS
maroonspot calicoflower	*Downingia concolor*	2026-05-15
narrowleaf milkweed	*Asclepias fascicularis*	2026-06-25
narrowleaf mule-ears	*Wyethia angustifolia*	2026-05-15
Narrowleaf Willow	*Salix exigua*	2026-06-24
needleleaf navarretia	*Navarretia intertexta*	2026-05-15
northern California black walnut	*Juglans hindsii*	2026-06-24
orange bush monkeyflower	*Diplacus aurantiacus*	2026-05-29
Oregon Ash	*Fraxinus latifolia*	2026-06-02
Pacific Bleeding Heart	*Dicentra formosa*	2026-05-04
Pacific madrone	*Arbutus menziesii*	2026-07-04
Pacific poison oak	*Toxicodendron diversilobum*	2026-06-02
pineapple-weed	*Matricaria discoidea*	2026-05-20
Redwood Sorrel	*Oxalis smalliana*	2026-05-28
small-headed clover	*Trifolium microcephalum*	2026-05-15
Superb Mariposa Lily	*Calochortus superbus*	2026-06-11
thimble clover	*Trifolium microdon*	2026-05-15
tomcat clover	*Trifolium willdenovii*	2026-05-15
Toyon	*Heteromeles arbutifolia*	2026-06-19
trailing blackberry	*Rubus ursinus*	2026-06-24
Tree Clover	*Trifolium ciliolatum*	2026-05-15
valley oak	*Quercus lobata*	2026-06-24
wavy-leafed soap plant	*Chlorogalum pomeridianum*	2026-05-20
western sycamore	*Platanus racemosa*	2026-05-17
white brodiaea	*Triteleia hyacinthina*	2026-05-15
white-tipped clover	*Trifolium variegatum*	2026-05-15
Winecup Clarkia	*Clarkia purpurea*	2026-05-15
Yellow Mariposa Lily	*Calochortus luteus*	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment Summary This site in the North Bay region supports a **mixed oak woodland and chaparral community**, dominated by blue oak and California black oak with understory elements of California buckeye, California bay, and drought-adapted shrubs like Phacelia and Dudleya. The species composition—particularly the prevalence of xeric-adapted plants (dry-site specialists) alongside riparian indicators like Braun's Giant Horsetail and box elder—suggests a **transitional zone between oak savanna and seasonal wetland or drainage areas**, likely with variable moisture availability throughout the year. Soils are probably well-drained in upland areas but may retain seasonal moisture in swales or near drainage paths, indicating that grading activities should avoid concentrated runoff that could destabilize slopes or alter natural drainage patterns to these wetter microsites. The presence of 57 species concentrated across 5,051 observations reflects a site of moderate ecological diversity that will require careful consideration of drainage design to prevent erosion or unwanted hydrologic changes that could degrade habitat quality. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

636	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants

VENOMOUS SNAKES

SPECIES	OBSERVATIONS
Crotalus oreganus oreganus	636 obs

VENOMOUS SPIDERS

SPECIES	OBSERVATIONS
Latrodectus hesperus	154 obs

STINGING INSECTS

SPECIES	OBSERVATIONS
Dasymutilla aureola	91 obs
Dasymutilla californica	91 obs
Dasymutilla sackenii	91 obs

TOXIC PLANTS

SPECIES	OBSERVATIONS
Toxicodendron diversilobum	1100 obs

SPECIES	OBSERVATIONS
Datura wrightii	57 obs
Datura stramonium	57 obs
Datura ferox	57 obs
Conium maculatum	199 obs
Urtica gracilis	83 obs
Urtica gracilis holosericea	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Northern California Survey Site **Highest-Priority Hazards:** Poison oak (Toxicodendron diversilobum) dominates this area with 1,439 observations and will be your primary concern—it's ubiquitous in Sonoma County scrub and woodland. Northern Pacific rattlesnakes (636 observations) are the second major threat, particularly in rocky, brushy terrain. Black widow spiders and various toxic datura species present secondary but serious risks. **Practical Avoidance:** Wear full-length pants, closed-toe boots, and long sleeves; learn to identify poison oak (three leaflets, waxy appearance) and avoid all contact, washing exposed skin immediately after work. Stay alert in tall grass and rocky areas where rattlesnakes shelter—make noise while moving, watch hand/foot placement during climbing or reaching, and never put hands in unseen spaces. Avoid touching or ingesting any unfamiliar plants, especially those with purple flowers or thorny seed pods. **Seasonal Considerations:** Rattlesnakes are most active April–October, with peak activity in summer heat; expect them to be less visible but still present during cooler months. Poison oak remains hazardous year-round but is most potent during growing season (spring/summer) when oils are concentrated. **Always verify current conditions locally before fieldwork.**

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3 Beam Paths

545ft Closest Beam

3 Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26 Total Airways

13 Jet Routes

2 Victor Airways

11 Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200	89	64	5	14	6
Total Observations	Total Species	Birds	Mammals	Reptiles	Amphibians

BIRDS (64 species)

COMMON NAME	*Scientific Name*	OBS
Wild Turkey	*Meleagris gallopavo*	8
Snowy Egret	*Egretta thula*	7
California Quail	*Callipepla californica*	5
Western Bluebird	*Sialia mexicana*	5
California Towhee	*Melozone crissalis*	5
Great Egret	*Ardea alba*	4
Red-tailed Hawk	*Buteo jamaicensis*	4
Black Phoebe	*Sayornis nigricans*	4
Northern Flicker	*Colaptes auratus*	3
Cooper's Hawk	*Astur cooperii*	3
Ash-throated Flycatcher	*Myiarchus cinerascens*	3
Western Screech-Owl	*Megascops kennicottii*	3
Turkey Vulture	*Cathartes aura*	3
Mallard	*Anas platyrhynchos*	3
Bushtit	*Psaltriparus minimus*	3

+ 49 additional species

MAMMALS (5 species)

COMMON NAME	*Scientific Name*	OBS
Coyote	*Canis latrans*	3
Black-tailed Jackrabbit	*Lepus californicus*	3
Mule Deer	*Odocoileus hemionus*	2
Eastern Fox Squirrel	*Sciurus niger*	1
Striped Skunk	*Mephitis mephitis*	1

REPTILES (14 species)

COMMON NAME	*Scientific Name*	OBS
Western Fence Lizard	*Sceloporus occidentalis*	10
Northern Pacific Rattlesnake	*Crotalus oreganus oreganus*	7
Pacific Gopher Snake	*Pituophis catenifer catenifer*	7
California King Snake	*Lampropeltis californiae*	3
Pond Slider	*Trachemys scripta*	2

COMMON NAME	*Scientific Name*	OBS
Western Skink	*Plestiodon skiltonianus*	2
Ring-necked Snake	*Diadophis punctatus*	1
Common Garter Snake	*Thamnophis sirtalis*	1
Western Yellow-bellied Racer	*Coluber constrictor mormon*	1
Pacific Ringneck Snake	*Diadophis punctatus amabilis*	1
Northwestern Pond Turtle	*Actinemys marmorata*	1
Common Sharp-tailed Snake	*Contia tenuis*	1
North American Racer	*Coluber constrictor*	1
Southern Alligator Lizard	*Elgaria multicarinata*	1

AMPHIBIANS (6 species)

COMMON NAME	*Scientific Name*	OBS
Pacific chorus frog	*Pseudacris regilla*	5
Western Toad	*Anaxyrus boreas*	3
American Bullfrog	*Lithobates catesbeianus*	3
Foothill Yellow-legged Frog	*Rana boylii*	3
Rough-skinned Newt	*Taricha granulosa*	1
California Giant Salamander	*Dicamptodon ensatus*	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment **Location: 38.3459, -122.3047 (Northern California)** This site supports a diverse wildlife community characteristic of oak woodland and grassland habitat transitioning to riparian or wetland areas, as evidenced by the substantial representation of waterfowl (egrets), aquatic reptiles (pond sliders, Northwestern pond turtles), and amphibians including sensitive species such as Foothill Yellow-legged Frog and Rough-skinned Newt. The presence of these amphibians and aquatic turtles indicates reliable seasonal or permanent water sources with supporting vegetation structure. Several species present warrant regulatory attention: Northern Pacific Rattlesnake, California King Snake, and Common Sharp-tailed Snake are sensitive to habitat fragmentation; Foothill Yellow-legged Frog is a California Species of Special Concern; and Northwestern Pond Turtle is declining range-wide and may require consultation under California environmental review. The moderate species diversity (89 species, 200 observations) and prevalence of both native predators (raptors, coyote) and generalist species (California Quail, towhees) suggest a landscape with mixed land uses experiencing moderate disturbance, with retained native vegetation patches supporting ecological function but likely fragmented or modified from pre-development conditions. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

☀ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = **38.3°** (latitude) → **9/12 pitch**

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	← excellent
7/12	30.3°	98.8%	← excellent
8/12	33.7°	99.3%	← excellent
9/12	36.9°	99.8%	← annual optimal
10/12	39.8°	99.8%	← excellent
12/12	45°	99%	← excellent

Going from 5/12 to 6/12 at framing costs ~$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~$108 value/yr). Over 25yr panel life: ~$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr·ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 · ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 · THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 · TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	⚠ verify
50ft	All year	4.8	4.6	7.3	⚠ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
⚠ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
†Balanced = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
⚠ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

⬡ AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Solar and Energy Planning for Your Home You're in a sweet spot for solar: your location gets about 6.6 peak sun hours daily on average, which is solid—think of it like having a reliable 6.6-hour workday where the sun is doing useful work on your panels, even though it's up much longer. Your roof pitch is nearly perfect for capturing that solar energy year-round, and here's where color matters hugely: a dark roof soaks up about 301 BTU of heat per hour per square foot in summer, which is like having a space heater running on your roof, but a light-colored roof cuts that in half—so that first choice saves you real money in cooling costs. Pay special attention to any windows on your west side of the house, because afternoon summer sun pouring through west glass adds over 100 BTU per hour per square foot, which is the peak heating problem when you're trying to cool down; consider shades or exterior blinds for those spots. Here's a smart and free move: a deciduous tree 40 feet away on your south side blocks about 4 hours of summer heat (keeping you cool when you need it) but drops its leaves in winter and lets that warming sun through (saving you heating energy when it's chilly). Thermal mass—think of it as the home's ability to absorb heat slowly and release it later, like a sponge for temperature swings—matters more with thicker walls, and your choice between standard framing versus thicker ICF blocks will affect how well your home naturally smooths out temperature changes. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

🧠 **SITE INTELLIGENCE SUMMARY** AI synthesis — Claude Sonnet

⚙ **Technical** 👤 **Client**

Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING **38.34586°N, 122.30466°W | Napa County, CA**

Structural Design Drivers & Climate Envelope

This site falls within IECC Climate Zone 3C (Marine) with an annual freezing degree-day accumulation of 0°F-days, indicating negligible frost risk and eliminating concern for seasonal ground heave or deep frost protection. Wind and snow load data are not available for this coordinate, which is atypical for a formal design submission and warrants clarification with the building official; however, the marine designation and zero AFI suggest a coastal or bay-influenced microclimate where wind loading may be the governing lateral design criterion rather than seismic demand. The absence of seismic hazard parameters in the dataset prevents determination of whether wind or seismic will control the primary structural system; this ambiguity must be resolved early through site-specific wind and seismic response studies before finalizing lateral load path design.

Geotechnical Foundation Strategy

The site is underlain by Coombs gravelly loam with 2 to 5 percent slopes, classified as Hydrologic Group C (moderately low infiltration) and well-drained texture. The well-drained classification supports shallow foundation design and eliminates expansive soil concerns common to fine-grained profiles. However, the "gravelly loam" matrix with Group C hydrology indicates the presence of impermeable layers or clay lenses that will govern subsurface drainage behavior; a Phase I geotechnical investigation should establish bearing capacity, settlement characteristics, and seasonal water table depth, particularly given the sloped topography which may concentrate perched water conditions. Frost depth is not applicable per zero AFI, so footing depth can be driven by bearing capacity and drainage requirements rather than frost protection.

Hazard Assessment

No Quaternary faults are present within the search radius, and no subsidence basin data exists for this location, both favorable conditions. The seismic hazard parameter is not populated in the dataset; given Napa County's proximity to the San Andreas system and regional fault networks, this omission is a critical gap that must be filled immediately through USGS seismic hazard maps and a Quaternary fault assessment within at least 15 kilometers. Wildfire risk data are not provided but should be verified against Cal Fire hazard zone maps, as Napa County contains areas of significant wildfire exposure.

Site Context & Mineral Activity

Three active mineral extraction operations exist within 5 miles: Basalt Rock Co (clay, 2 miles), Mt George Pumice Deposit (3.1 miles), and Silva Pumice Quarry (5 miles). While no active oil and gas wells are operating within 10 miles—the nearest being a plugged Mobil Oil well 7.2 miles distant—the proximity of active quarries warrants confirmation that blasting, dust, and ground vibration from extraction will not affect building performance or occupancy use. Eight historical oil and gas wells are recorded within the 10-mile radius; their abandonment status should be cross-checked to eliminate subsurface hydrocarbon contamination or subsidence risks.

First Priority Investigation

Obtain site-specific seismic hazard parameters (spectral accelerations, fault proximity, soil amplification) and clarify wind load criteria for Zone 3C before proceeding to lateral system selection.

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SITE SUMMARY

Coordinates	38.345860°N 122.304660°W
Elevation	66 ft NAVD88 (approx)
Township/Range	T6N, R4W, Mount Diablo Meridian
FEMA Zone	Zone X Panel 06055C0510FF
Nearest BM	JT9010 0.2mi 62.7ftft
Nearest CORS	P261 14.1mi
State Plane	CA 2 (0402)

TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

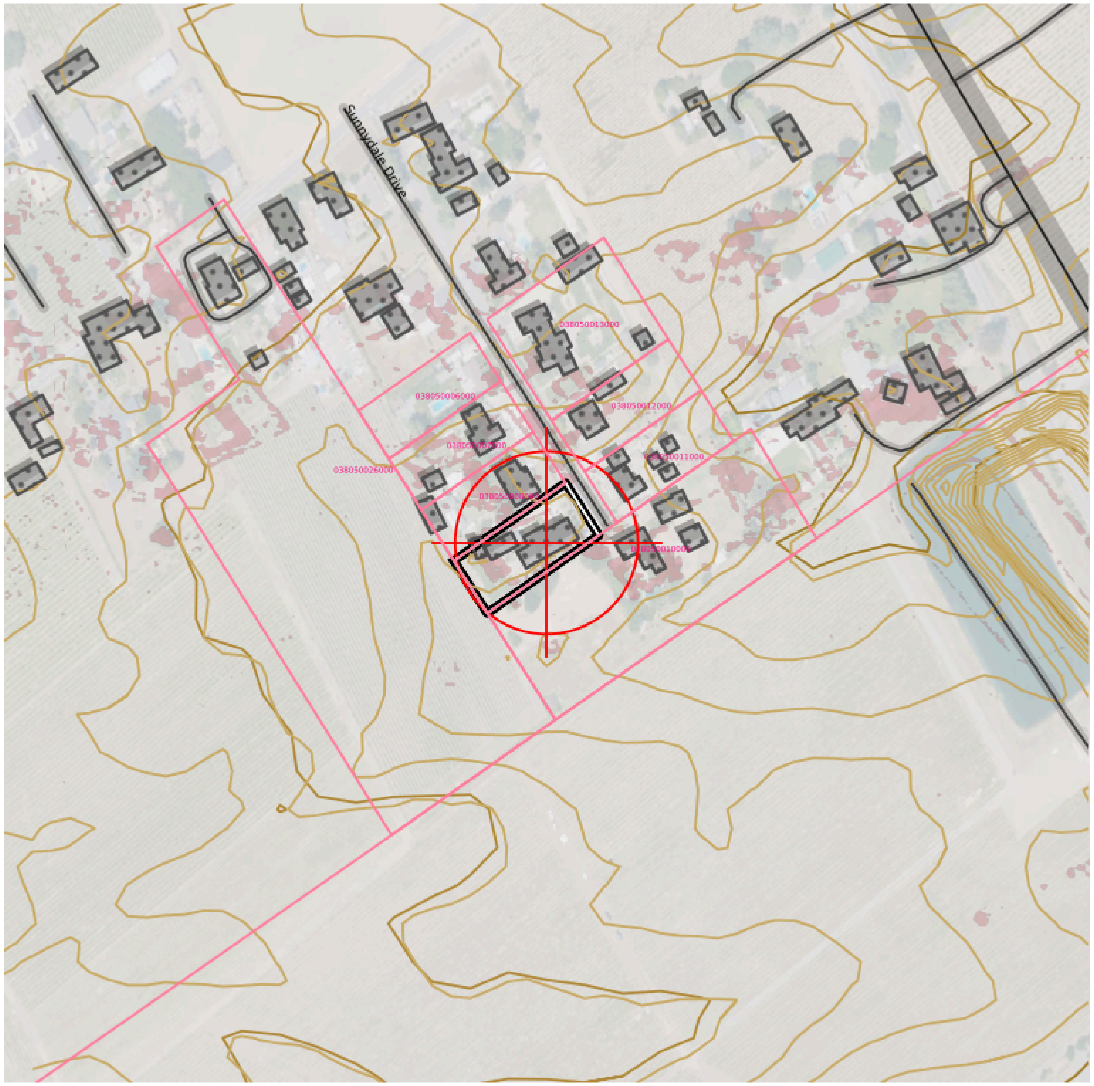
Generated: July 13, 2026
Base: USGS National Map
Generated by TopoScout · outKastdesigns.com

TOPOSCOUT SITE DXF PREVIEW

45 CAD Layers · AutoCAD Compatible · State Plane Coordinates

Need the editable CAD file? Add the DXF at toposcout.com

TOPOSCOUT SITE PLAN PREVIEW



INCLUDED LAYERS

N

0 —— 400 ft State Plane · NAD83 · US Feet

■ Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels ■ County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polylines (all geometry)

SITE AERIAL — 600 ft × 600 ft

38.34586°N, 122.30466°W · Google Aerial

TOPOSCOUT SITE AERIAL IMAGERY

Source: Google Aerial

AREA CONTEXT — 3000 ft × 3000 ft

38.34586°N, 122.30466°W · Google Aerial

⚡ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online – https://trimblegnssplanningonline.com/
Satellite visibility, DOP charts, sky plots

FAA B4UFLY – https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly
Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – https://www.aloft.ai/
Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp – https://www.swpc.noaa.gov/products/planetary-k-index
Real-time Kp index, geomagnetic storm alerts

BASE STATION & NETWORK

NOAA CORS Network – https://geodesy.noaa.gov/CORS/cors-data.shtml
Continuously operating reference stations, data download

NGS CORS Station Map – https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml
Interactive map of all CORS stations with status

IGS Station Monitor – https://www.igs.org/network
International GNSS Service – global base station health

POST-PROCESSING

NGS OPUS – https://geodesy.noaa.gov/OPUS/index.jsp
Online Positioning User Service – static GNSS processing

NGS OPUS-RS – https://geodesy.noaa.gov/OPUSI/index.jsp
Rapid static processing – sessions as short as 15 minutes

CORS Data Download – https://geodesy.noaa.gov/CORS/cors-data.shtml
Raw RINEX observation files from CORS stations

REFERENCE & CALCULATIONS

NGS Datasheets – https://www.ngs.noaa.gov/cgi-bin/datasheet.prl
Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) – https://geodesy.noaa.gov/NCAT/
NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination – https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml
Declination calculator for any location and date

NGS Geoid Models – https://geodesy.noaa.gov/GEOID/
GEOID18 and current model downloads, GEOID calculator

FEMA & FLOOD

FEMA Flood Map Service Center – https://msc.fema.gov/portal/home
FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate
Current EC form, instructions, LOMA/LOMR submission guidance

DRONE OPERATIONS

FAA Part 107 — Commercial Rules – https://www.faa.gov/uas/commercial_operators
Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – https://faadronezone.faa.gov/
Drone registration, operator certificates, waiver management

FAA B4UFLY – https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly
Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers – https://www.faa.gov/uas/commercial_operators/part_107_waivers
Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – https://www.aloft.ai/
Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

AirNow Fire & Smoke Map
https://www.fire.airnow.gov
Current fire locations, smoke plumes, PM2.5 readings by location

AirNow Interactive Map
https://gispub.epa.gov/airnow
Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

NOAA Smoke Forecast
https://airquality.weather.gov
48-hr smoke concentration forecast, HRRR-Smoke model

Links verified: February 2026 · Check quarterly · TopoScout GNSS Planning System